DS	**Session**	Stack	**Question Information**	Level 1 Challenge 47

Question description

You are given an array *A* of *N* integers. Now, two functions $F(X)$ and $G(X)$ are defined:

$F(X)$: This is the smallest number *Z* such that $X < Z \leq N$ and $A[X] < A[Z]$

$G(X)$: This is the smallest number *Z* such that $X < Z \leq N$ and $A[X] > A[Z]$

Now, you need to find for each index *i* of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index *i*, output *1* as its answer. If such a number does exist, output $A[G(F(i))]$

Constraints:

1 ≤ N ≤ 3000

0 ≤ A[i] ≤ 10^18

Input :

The first line contains a single integer *N* denoting the size of array *A*. Each of the next *N* lines contains a single integer, where the integer on the ith line denotes $A[i]$.

Output :

Print *N* space-separated integers on a single line, where the ith integer denotes $A[G(F(i))]$ or *1*, if $G(F(i))$ does not exist.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
8
3 7 1 7 8 4 5 2
```

EXPECTED OUTPUT

```
1 4 4 4 -1 2 -1 -1
```

Test Case 2

INPUT (STDIN)

```
8
31 71 11 17 81 14 51 21
```

EXPECTED OUTPUT

```
11 14 14 14 -1 21 -1 -1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
if(a[j]>a[i])
```

Test Case 2

KEYWORD

```
for(i=0;i<n;i++)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
9
```

Test Case 2

TOKEN COUNT

```
225
```

Test Case 3

NLOC

```
30
```

Code

solution.c

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (int i = 0; i < n; i++) scanf("%d", &a[i]);
    for (int i = 0; i < n; i++) {
        int f = -1;
        for (int j = i + 1; j < n; j++) {
            if (a[j] > a[i]) { f = j; break; }
        }
        if (f == -1) {
            printf("-1%s", (i == n - 1) ? "" : " ");
            continue;
        }
        int g = -1;
        for (int k = f + 1; k < n; k++) {
            if (a[k] < a[f]) { g = k; break; }
        }
        if (g == -1) printf("-1%s", (i == n - 1) ? "" : " ");
        else printf("%d%s", a[g], (i == n - 1) ? "" : " ");
    }
    printf("\n");
    return 0;
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
8
3 7 1 7 8 4 5 2
1 4 4 4 -1 2 -1 -1
PS C:\C Programming>
```